Medical Discharge Summary Simplification for Patients

BTP Project Report

May 9, 2026

1 Introduction

The objective of this project is to automate the simplification of complex medical discharge summaries into plain "Lay English" that can be easily understood by patients and their families. Medical documents are often laden with technical jargon, abbreviations, and complex diagnostic codes (ICD codes) that pose a significant barrier to patient understanding. This project develops a robust pipeline to bridge this communication gap using fine-tuned Large Language Models (LLMs).

Patient comprehension of discharge instructions is a critical determinant of post-hospitalization outcomes. Studies have consistently shown that patients who understand their discharge summaries are more likely to adhere to follow-up care, take medications correctly, and recognize warning signs that require urgent attention. Despite this, discharge summaries are routinely written for clinical audiences, not patients or their families. Automating lay language conversion at scale presents a meaningful opportunity to improve healthcare equity and patient safety.

2 Task Description

The task involves a sequence-to-sequence transformation where the input is a technical medical discharge summary and the output is a simplified version that preserves all critical clinical information while using accessible language. Key challenges include:

- **Information Preservation:** Ensuring no clinical details, medications, or surgical events are lost during simplification.
- **Domain Specificity:** Handling highly specialized medical terminology, including drug names, surgical procedures, and laboratory values.
- **Document Length:** Processing long discharge summaries that exceed the token limits of standard transformer models.
- **Factual Fidelity:** Avoiding hallucinations — the generation of plausible-sounding but factually incorrect medical statements — which can be dangerous in a clinical communication context.
- **Tone and Sensitivity:** Handling emotionally sensitive content such as end-of-life events, death declarations, and deteriorating prognoses with appropriate language while remaining clinically accurate.

- **Structural Coherence:** Maintaining the logical flow of a clinical narrative across multiple conditions and events without merging, reordering, or omitting distinct clinical episodes.

3 Datasets

3.1 Fine-tuning Dataset: `data.tsv`

The models are fine-tuned on the `data.tsv` dataset, which contains parallel pairs of original medical texts and their simplified English counterparts. To ensure the model learns simplification rather than aggressive summarization, a preservation filter is applied to discard samples where the simplified text is significantly shorter than the original (ratio < 0.45). Due to computational resource constraints on the Kaggle platform—specifically GPU memory and session time limits—fine-tuning was restricted to a subset of 10,000 samples.

3.2 Inference Dataset: `anotated_dataset_v2.xlsx`

The final evaluation is performed on the `anotated_dataset_v2.xlsx` dataset. This contains real-world discharge summaries along with patient outcomes and ICD-10 codes. The pipeline expands abbreviations and applies a medical lay-dictionary before passing the text to the fine-tuned LLM.

4 Methodology

The pipeline consists of several stages:

1. **Preprocessing:** Normalization and expansion of abbreviations.
2. **Lay-Dictionary Mapping:** Replacing known medical terms with simplified Indian English equivalents (e.g., "hypertension" → "high BP").
3. **Sliding Window Chunking:** Since most models have a 512 or 1024 token limit, long summaries are split into overlapping chunks (450 tokens with 50-token overlap) to ensure continuity and prevent information loss at boundaries.
4. **Chunked Generation:** Each chunk is processed independently by the model, and the results are stitched together to form the final document.

5 Models Explored

5.1 BioGPT

Architecture: BioGPT follows the Transformer decoder-only backbone, specifically adopting the GPT-2 medium configuration with 24 layers, a hidden size of 1024, and 16 attention heads. The final model contains approximately 347 million parameters.

Pre-training Data: Unlike models that undergo continuous pre-training, BioGPT was pre-trained from scratch on a massive dataset of 15 million PubMed abstracts (including both titles and abstracts). A specialized in-domain vocabulary was constructed using Byte Pair Encoding (BPE) on this corpus, resulting in a vocabulary size of 42,384, which is highly optimized for biomedical terminology.

Why BioGPT for this task:

- **Generative Excellence:** As a causal LM, it is highly effective at generating fluent, natural language.
- **From-Scratch Domain Expertise:** Pre-training from scratch on PubMed allows the model to learn biomedical semantics without the interference of general-domain biases.
- **State-of-the-Art Performance:** As demonstrated in Luo et al. (2022), BioGPT achieves superior performance on various biomedical generation and mining tasks compared to general-purpose GPT models.

Fine-tuned Checkpoint: `https://huggingface.co/11Raghav/BioGPT`

5.2 SciFive

Architecture: SciFive is a domain-specific model based on the T5 (Text-to-Text Transfer Transformer) encoder-decoder architecture. It was initialized from the pre-trained weights of the base T5 model and then specifically adapted for the biomedical domain.

Pre-training Data: As detailed in Phan et al. (2021), SciFive underwent extensive re-training for an additional 200,000 steps (for the base version) on a combination of the general-domain C4 corpus, PubMed abstracts, and the PubMed Central (PMC) full-text corpus. This multi-corpus approach ensures the model retains general linguistic knowledge while mastering dense clinical and technical narratives.

Why SciFive for this task:

- **Text-to-Text Versatility:** By treating simplification as a text-to-text problem (similar to translation), SciFive is inherently more flexible than encoder-only models like BERT.
- **Complex Output Handling:** SciFive has demonstrated superior performance on biomedical tasks requiring long, coherent outputs (e.g., BioASQ question answering), making it well-suited for the detailed rewriting required in discharge summary simplification.
- **PMC Full-Text Advantage:** Pre-training on full PMC articles, rather than just abstracts, provides the model with better context for the narrative flow found in medical reports.

Fine-tuned Checkpoint: `https://huggingface.co/11Raghav/SciFive`

5.3 BioBART

Architecture: BioBART is based on the BART (Bidirectional and Auto-Regressive Transformers) architecture. It utilizes a bidirectional encoder and an auto-regressive decoder, making it highly effective for sequence-to-sequence tasks.

Pre-training Data: As detailed in Yuan et al. (2022), BioBART is initialized from the general-purpose BART-base/large checkpoints and continuously pre-trained on the PubMed abstracts corpus (approx. 41 GB of text). A key technical finding in the paper is that using only the *text-infilling* task for adaptation yields better performance on biomedical NLG tasks than including sentence permutation.

Why BioBART for this task:

- **Seq2Seq Mastery:** The encoder-decoder nature of BART is specifically designed for rewriting and summarization, which maps directly to the simplification goal.
- **Domain Adaptation:** Its continuous pre-training on biomedical corpora allows it to maintain medical entity integrity while generating fluent English.

Fine-tuned Checkpoint: `https://huggingface.co/11Raghav/BioBART`

5.4 Flan-T5

Architecture: Flan-T5 is an instruction-tuned version of the T5 (Text-to-Text Transfer Transformer) encoder-decoder architecture. The version used in this project, Flan-T5-base, contains approximately 250 million parameters.

Pre-training Data: As described by Chung et al. (2022), Flan-T5 was fine-tuned on the FLAN (Finetuned Language Net) collection, which was scaled to 1,836 tasks across 473 datasets and 146 task categories. A critical component of its training was the inclusion of *chain-of-thought* (CoT) data, which significantly improves the model's reasoning and instruction-following performance on unseen tasks.

Why Flan-T5 for this task:

- **Instruction Scaling:** The massive variety of instruction-based tasks in its pre-training allows Flan-T5 to generalize exceptionally well to specific simplification prompts.
- **Robustness:** It provides strong zero-shot and few-shot performance, often outperforming much larger models that lack instruction tuning.
- **Reliability:** Compared to ClinicalT5, Flan-T5 demonstrated superior stability in following the "preserve every detail" constraint without degenerating into repetitive output.

Fine-tuned Checkpoint: `https://huggingface.co/11Raghav/FlanT5`

5.5 Pegasus-PubMed

Challenges: Pegasus-PubMed was explored due to its specialized architecture for abstractive summarization. However, it was found to be unsuitable for the Kaggle environment in this project.

Architecture & Pre-training: As detailed in Zhang et al. (2020), Pegasus utilizes a standard Transformer encoder-decoder architecture with approximately 568 million parameters (in the large version). Its defining feature is the *Gap Sentences Generation* (GSG) pre-training objective, where entire important sentences are masked from a document and the model is tasked with generating them. This objective is designed to narrow the gap between pre-training and the downstream task of abstractive summarization.

Why it failed in this context:

- **Memory Intensity:** Due to its large parameter count and the complexity of the GSG-informed attention mechanism, the model consistently triggered `OutOfMemoryError` (OOM) on the Kaggle T4 x2 GPU environment.
- **Input Scaling:** The model's efficiency on long clinical documents was limited by the high VRAM requirements of its specific Transformer implementation when processing documents at the 512-1024 token scale.

5.6 ClinicalT5

Challenges: ClinicalT5 was initially a primary candidate given its specialization in clinical notes. However, it was ultimately discarded due to technical failures during inference.

- **Output Degeneration:** The model suffered from a severe "boundary bug" during chunked generation. When input token sequences were stitched together, the SentencePiece tokenizer's word-boundary markers (▁) were corrupted, causing the model to degenerate and output repeating characters (e.g., "sssss").
- **Instruction Fragility:** Compared to instruction-tuned models like Flan-T5, ClinicalT5 was less robust at following the complex simplification prompts, often resulting in outputs that were inaccurate.

6 Quantitative Evaluation: Readability and Compression

To objectively measure the performance of the four fine-tuned models, we conducted a quantitative analysis on a test set of 200 medical discharge summaries. The evaluation focused on two key dimensions: **Readability**, measured using the Flesch Reading Ease score, and **Compression**, measured by the change in word count and the compression ratio.

The Flesch Reading Ease score indicates how easy a text is to understand; higher scores indicate material that is easier to read. For context, a score of 60-70 is considered standard English, while scores below 30 are typical of highly technical or academic documents. The original medical summaries in our test set had an average Flesch score of **20.24**, reflecting their high technical complexity.

Table 1 summarizes the results across all models.

Table 1: Quantitative comparison of model performance on 200 summaries.

Model	**Avg. Flesch**	**Flesch Improv.**	**Avg. Words**	**Comp. Ratio**
Original Text (Baseline)	20.24	—	598.7	1.000
BioGPT	**61.77**	**+41.53**	**567.9**	**0.946**
SciFive	53.61	+33.37	668.2	1.108
BioBART	47.32	+27.08	656.1	1.086
Flan-T5	12.69	-7.55	781.7	1.327

6.1 Analysis of Quantitative Results

- **BioGPT as the Top Performer:** BioGPT achieved the highest average Flesch score (61.77) and the greatest improvement (+41.53). It was the only model that consistently reduced the word count (compression ratio of 0.946), suggesting a more efficient simplification process that avoids redundant technical detail.
- **SciFive and BioBART:** Both SciFive and BioBART showed significant improvements in readability. However, they both tended to *expand* the text (ratios > 1.0), likely due to their tendency to add explanatory phrases or maintain more of the original clinical structure.
- **Flan-T5 Failure:** Flan-T5 performed poorly in this quantitative evaluation, actually reducing the average Flesch score to 12.69. This is consistent with our qualitative observation that Flan-T5 often copied the original text verbatim or introduced garbled content, both of which negatively impact readability metrics.
- **Universality of Improvement:** While BioGPT, SciFive, and BioBART improved readability in nearly all cases, the qualitative failures (hallucinations) noted in Section 7 remain a critical concern that these automated metrics do not capture.

7 Qualitative Evaluation on a Representative Example

7.1 Input Discharge Summary

To qualitatively evaluate and compare the four fine-tuned models, we present their outputs on a representative discharge summary involving a complex, multi-morbidity case with a fatal outcome. This example is particularly informative because it tests the model's ability to handle: multiple simultaneous diagnoses, escalating clinical deterioration, critical interventions (resuscitation, cardioversion, dialysis), and a death declaration — all of which require precise, sensitive, and factually accurate lay language conversion.

Original Discharge Summary:
admission: 12/06, discharge: 12/21, stay: 14 d 11 h, location: ward, payer: sus. 1. heart failure profile c dobutamine on 12/07 and 12/08, 2. abscess in the left thigh, osteomyelitis in the pubis, fournier's syndrome, drainage on 12/03 with ultrasound and drain placement on 12/15, treatment with ceftriaxone guided by culture of abscess escherichia coli multi s colonized antibiotic therapy after shock, incision in intensive care unit on 12/17 with abundant purulent secretion, {omitted}. 3. septic shock from cutaneous urinary focus on 12/16 polymyxin d0 12/15 amikacin and vancomycin on 12/17, 4. chronic kidney disease acute dialysis by fistula on 12/16 5. cardiorespiratory arrest with pulseless electrical activity, 10 minutes after intubation report of previous bradycardia, follow-up appointment after 1 minute of cardiopulmonary resuscitation, 6. vt with pulse after cardiorespiratory arrest, effective electrical cardioversion with a shock. [person] presenting with instability in the last half hour, requiring norepinephrine elevation up to 1.5 mcg/kg/min. progresses with bradycardia, qrs widening, and worsening shock. given the hyperkalemia even after hemodialysis, 2 ampoules of calcium gluconate and 100 ml of bicarbonate were administered, and the respiratory rate was increased on the ventilator in order to reduce pco2 and consequently k. 2 ampoules of atropine were administered without benefit. [person]

progressed to asystolic cardiorespiratory arrest, {omitted} in refractory shock. i declare {omitted} dead, due to absence of central pulses, pupils in fixed mydriasis, absence of heart sounds and asytole on the monitor, as well as absence of brainstem reflexes. the body was released. i request contact with the family. i await the death certificate to be completed. death. drainage of fournier's gangrene. death.

The following subsections present each model's output, followed by a detailed error analysis.

7.2 Reference: Expected Lay Language Summary

For reference, the key clinical facts that a correct lay language conversion must preserve and accurately communicate are:

1. Admission on December 6, discharge (death) on December 21, 14-day stay.

2. Heart failure treated with dobutamine (a medication to support heart pumping) on December 7 and 8.

3. A pus-filled abscess in the left thigh, bone infection (osteomyelitis) in the pubic bone, and Fournier's gangrene (a severe, life-threatening infection of the genital/perineal area). Drainage performed on December 3 via ultrasound, with a drain placed on December 15. Antibiotic ceftriaxone was used based on bacterial culture results identifying *Escherichia coli*. A surgical incision was performed in the ICU on December 17.

4. Septic shock (life-threatening infection spreading to the bloodstream) originating from a skin/urinary source on December 16. Treated with polymyxin (from December 15), amikacin, and vancomycin (from December 17).

5. Chronic kidney disease requiring emergency dialysis via a fistula on December 16.

6. Cardiac arrest (heart stopped) with no electrical pulse detectable, occurring 10 minutes after a breathing tube was inserted. The patient was resuscitated with CPR (chest compressions) for 1 minute.

7. Ventricular tachycardia (a dangerous fast heart rhythm) after resuscitation, successfully treated with an electrical shock (cardioversion).

8. Final deterioration: dangerously low blood pressure requiring high-dose norepinephrine (a medication to raise blood pressure), abnormal heart rhythm on the monitor (QRS widening), and severely elevated potassium in the blood (hyperkalemia) resistant to dialysis. Calcium gluconate (2 vials) and bicarbonate (100 ml) were given to protect the heart; ventilator breathing rate was increased to lower potassium indirectly. Atropine (2 vials) was given to treat slow heart rate without effect.

9. The patient progressed to a final, irreversible cardiac arrest (asystole). Death was declared based on absence of central pulses, fixed dilated pupils (mydriasis), absence of heart sounds, flatline on monitor, and absence of brainstem reflexes. The family was contacted and the death certificate was completed.

7.3 BioBART Output and Error Analysis

BioBART Output:
This report describes a patient who was admitted to the hospital on 12/06. She had several medical problems, including heart failure, osteomyelitis, and fournier's syndrome. She was treated with ceftriaxone guided by a culture of the swelling, antibiotics for septic shock, and incision in the intensive care unit (ICU). She also had septic shock from a focus in the urinary system on 12/07 and 12/08. She also had chronic kidney disease (chronic kidney disease) on 12/15. She had a sudden drop in her heart rate 10 minutes after being intubated, and she was able to return to her doctor after 1 minute of cardiopulmonary regovernment health schemecitation. The report also showed that the patient was being treated with calcium gluconate and bicarbonate, and the respiratory rate was increased to reduce pco2 and heart rate. She was also treated with atropine, a muscarinic receptor antagonist. The report concludes that this patient is a very rare case of death, and that it's important to carefully consider the patient's wishes when treating this serious condition. The patient was released from the hospital and contacted with the medical team. This case is unusual because the body was released from the hospital. The patient was able to communicate with the family, but the death certificate had to be completed. The death was caused by fournier's gangrene, a type of skin disease that affects the skin.

Errors and Limitations:

- **Incorrect gender assignment:** The original summary contains no gender information, yet BioBART incorrectly refers to the patient as "she" throughout. This is a hallucination that introduces inaccurate personal information.
- **Wrong dates for septic shock:** The septic shock occurred on 12/16, but BioBART incorrectly states it occurred on "12/07 and 12/08" — dates that correspond to the dobutamine treatment for heart failure. This is a critical factual error caused by cross-contamination of information across clinical events.
- **Wrong date for kidney disease:** The acute dialysis for chronic kidney disease occurred on 12/16, but the model states "12/15," which was actually the date of drain placement for the abscess. Again, dates are confused across clinical events.
- **Abscess omitted:** The pus-filled abscess in the left thigh — a primary source of infection — is not mentioned in the output. This is a significant omission of a clinically important finding.
- **Garbled text ("regovernment health schemecitation"):** The phrase describing cardiopulmonary resuscitation (CPR) is rendered as meaningless text. This reflects a tokenization or decoding error, likely occurring at a chunk boundary, producing output that is not only unintelligible but potentially alarming to a patient's family.
- **Incorrect explanation of respiratory rate increase:** The model states breathing rate was increased "to reduce pco2 and heart rate." The actual clinical goal was to reduce pCO2 in order to lower elevated blood potassium (hyperkalemia) — not to reduce heart rate. This is a clinically dangerous misinterpretation.
- **Atropine described as a "muscarinic receptor antagonist":** Instead of simplifying the term, BioBART introduces a more technical pharmacological descriptor. This is the opposite of the intended lay language conversion and is unsuitable for a patient-facing document.

- **Death trivialized and misrepresented:** The model describes the death as "a very rare case" and states "the patient was released from the hospital and contacted with the medical team" — implying the patient was alive and discharged. This is a severe misrepresentation of the clinical outcome. Additionally, characterizing Fournier's gangrene as merely "a type of skin disease that affects the skin" grossly understates the severity of a life-threatening necrotizing infection.

- **Missing critical interventions:** Dobutamine infusion, polymyxin, amikacin, vancomycin, calcium gluconate dose, bicarbonate dose, electrical cardioversion, and the CPR timeline are either omitted or inaccurately described.

Overall Assessment: BioBART's output is factually unreliable, contains hallucinated content, confuses dates across clinical events, and catastrophically misrepresents the patient's death. It is not suitable for lay language conversion of discharge summaries in its current fine-tuned state.

7.4 BioGPT Output and Error Analysis

BioGPT Output:
This text is about a patient who was admitted to the hospital. She was sent home from the hospital and stayed in the hospital for an average of 14 days. She was diagnosed with heart failure, a condition where the heart pump doesn't work properly. She also had a swelling in her left thigh, bone infection, and a condition called Fournier's syndrome. She was treated with a medicine called ceftriaxone, which is a type of antibiotic. She also received a medicine called polymyxin d0, a type of medication that helps kill bacteria. She was given a medicine called vancomycin, which is used to treat infections in the urinary tract. She was also treated with a diuretic called calcium gluconate and bicarbonate. She recovered from the infection and was able to breathe normally. However, she also had problems with her heart rate, blood pressure, and breathing. This case highlights the importance of being aware of potential problems when treating patients, especially when they are in a critical condition. It also highlights the need for close communication between the patient and their family to ensure they receive the best possible care. The authors recommend that the patient's family be involved in the decision-making process to ensure the best possible treatment. This is a case report of a patient who died from a condition called "driasis," which is a condition where the body doesn't have enough blood flow. The patient didn't have any heart sounds or any movement on their monitor, and they didn't show any signs of movement in the brain. This is a rare case of a patient dying from [truncated]

Errors and Limitations:

- **Incorrect gender assignment:** As with BioBART, BioGPT incorrectly assigns a female gender ("she") to a patient whose gender is not specified in the source text — an unprompted hallucination.

- **Contradictory statements about outcome:** The model first states "she recovered from the infection and was able to breathe normally," then later states the patient died. These two statements are directly contradictory within the same output. A patient's family reading this would be confused and misled.

- **Hallucinated condition "driasis":** The model invents a fictional medical condition called "driasis," described as "a condition where the body doesn't have enough blood flow.” This is a hallucination — the original text refers to “fixed mydriasis” (fixed dilation of the pupils, a sign of brain death), which is a clinical finding, not a cause of death.

- **Incorrect description of calcium gluconate:** The model refers to calcium gluconate as “a diuretic” (a medication that increases urine output). Calcium gluconate is actually a cardiac-protective agent used to counteract the effects of hyperkalemia on the heart. This is a serious factual error that could cause misunderstanding of the patient’s treatment.
- **Incorrect description of vancomycin:** The model states vancomycin is “used to treat infections in the urinary tract.” While vancomycin is a broad-spectrum antibiotic, in this case it was used for the treatment of septic shock and bacterial infection — not specifically a urinary tract infection. The simplification introduces a misleading specificity.
- **Output truncated mid-sentence:** The output ends abruptly with “This is a rare case of a patient dying from” — the generation was cut off, likely due to a token limit being reached. This means the model failed to complete the simplification of the full document.
- **Discharge summary written as a “case report”:** BioGPT reframes the output as an academic case report (“The authors recommend...”), adopting a writing style completely inappropriate for a patient-facing document. This is a domain confusion failure, likely arising from its pre-training on PubMed literature.
- **Missing critical clinical details:** The abscess drainage procedures, the drain placement date, the ICU incision, amikacin, norepinephrine dosing, CPR details, QRS widening, hyperkalemia management, atropine, bicarbonate, cardioversion, and the death declaration criteria are all absent from the output.
- **Inappropriate moralizing:** The output inserts unsolicited commentary about “the importance of being aware of potential problems” and family involvement in decision-making. These additions are not present in the source text and are inappropriate for a factual discharge summary conversion.

Overall Assessment: BioGPT conflates the task of lay language simplification with that of writing a biomedical case report, which is a direct consequence of its exclusive pre-training on PubMed abstracts. It hallucinates conditions, contradicts itself on patient outcome, and truncates its output. It is not suitable for this task in its current state.

7.5 Flan-T5 Output and Error Analysis

Flan-T5 Output:
admission: 12/06, sent home from hospital: 12/21, stay: 14 d 11 h, location: ward, payer: sus. 1. heart failure profile c dobutamine on 12/07 and 12/08, 2. pus-filled swelling in the left thigh, osteomyelitis in the pubis, fournier’s syndrome, drainage on 12/03 with ultrasound and drain placement on 12/15, treatment with ceftriaxone guided by culture of pus-filled swelling escherichia coli multi s colonized infection-fighting medicine therapy after shock, incision in ICU (serious care room) on 12/17 with abundant purulent secretion, omitted 4. chronic kidney disease acute dialysis by fistula on 12/16 5. cardiorespiratory arrest with

pulseless electrical activity, 10 minutes after intubation report of previous slow heartbeat, return visit to doctor appointment after 1 minute of cardiopulmonary resuscitation, 6. vt with pulse after cardioreaspirary arrest, effective electrical cardioversion with a shock. [...] Given the Hyperkalemia even following kidney diazepam, 2 ampuels of calcium Gluconate were administered [...] 3 ampoules were given without benefit. [person] progressed with asysTOLIC cardiorespritory arrest, Omitted in Refractory Shock. I Declare Omitted Dead, Due to Abstinence. [...] Simplify the following medical discharge summary into plain English that a patient’s family can understand [...] Death. Drainage of fournier’s gangrene. Death. , . , . , . . ’ ,’. ’,’. ” ; : . ; . : .. [truncated]

Errors and Limitations:

- **Near-verbatim copying of source text:** Flan-T5 largely reproduces the original discharge summary with only superficial substitutions (e.g., “abscess” → “pus-filled swelling”). This is not lay language conversion — it is near-copy output. The structural complexity, clinical notation format, and numbered list style of the original are preserved entirely, failing the core task objective.

- **Hallucinated drug “kidney diazepam”:** The model generates the phrase “kidney diazepam” as a substitution for hemodialysis (kidney dialysis). Diazepam is a sedative medication with no clinical relevance to dialysis or potassium management. This is a dangerous hallucination that introduces a completely incorrect medication name into a clinical record.

- **Incorrect ampoule count:** The original text states 2 ampoules of atropine were administered. Flan-T5 changes this to “3 ampoules,” an incorrect factual alteration of a medication dosage, which is clinically significant.

- **Death cause changed to “Abstinence”:** The death declaration states the cause as absence of central pulses, fixed pupils (mydriasis), and absence of brainstem reflexes. Flan-T5 replaces this entirely with “Due to Abstinence” — a completely fabricated and medically nonsensical cause of death. This is perhaps the most dangerous error in any of the model outputs, as it directly falsifies the documented cause of death.

- **Prompt leakage into output:** The model reproduces its own instruction prompt (“Simplify the following medical discharge summary into plain English...”) verbatim within the generated output. This is a severe instruction-following failure indicating the model could not distinguish between input instructions and output content.

- **Garbled punctuation cascade:** The output ends with a long sequence of incoherent punctuation marks and whitespace characters (, . , . , . . ’ , ’. ’, ’. ’’ ; : . ; .). This is characteristic of a decoder entering a degenerate repetition loop, likely triggered by the chunked boundary stitching process.

- **Inconsistent capitalization and spelling:** The output contains erratic capitalization (“asysTOLIC,” “cardiorespritory,” “Hyperkalemia”) and spelling errors (“ampuels” instead of “ampoules,” “cardioreaspirary” instead of “cardiorespiratory”). These likely arise from the model reproducing partial tokens from overlapping chunks without coherent decoding.

- **Item 3 (septic shock) omitted:** The entire septic shock event — including polymyxin, amikacin, and vancomycin — is missing from the output. Given that septic shock was the direct precipitant of the patient's deterioration and death, this is a critical omission.

- **No prose conversion achieved:** Despite being instruction-tuned, Flan-T5 fails to convert the bullet-point clinical notation into flowing, patient-readable prose. The output retains the numbered list structure and terse clinical abbreviations of the original.

Overall Assessment: Flan-T5 exhibits severe instruction-following failure in this domain, including prompt leakage, near-verbatim copying, dangerous drug hallucinations, falsification of cause of death, and decoder degeneration. Its fine-tuning on this medical dataset appears to have interfered with its instruction-following capabilities. It is not suitable for lay language conversion.

7.6 SciFive Output and Error Analysis

SciFive Output:
This patient was admitted to the hospital on 12/06 and was sent home from the hospital on 14/21. The hospital stay was 14 days and 11 hours. The patient had a history of heart failure and was given dobutamine on 12/07 and 12/08, a medicine called dobutamine, a swelling in the left thigh, osteomyelitis in the pubis, and a condition called fournier's syndrome. The patient was treated with ultrasound and antibiotics, and the patient was sent to the intensive care unit (ICU) on 12/17 with a lot of pus in the swelling. He was treated with ceftriaxone based on the culture of the pus. The patient developed septic shock from a skin focus on 12/16, which was treated with polymyxin d0 12/15 and amikacin and vancomycin, and chronic kidney disease acute dialysis by fistula on 12/16. The patient also had a slow heartbeat, widening of the chest, and worsening shock. Because of the high potassium levels in the kidneys, the patient was given calcium gluconate and bicarbonate, and their breathing rate was increased on the ventilator to reduce pCO2. The patient did not receive any benefit from atropine. The patient progressed to asystolic cardiorespiratory arrest, omitted in refractory shock. When the person was released from the hospital without benefit, they went into a serious condition called cardiorespiratory arrest. This was a condition where the heart and lungs were blocked, and the patient died. The person was released and the body was resuscitated. The body was released. The family requested contact with the family and the family requested the death certificate to be completed. The death was due to a breakdown of the gangrene, which was caused by a burn in the ear, nose, and throat.

Errors and Limitations:

- **Fabricated discharge date:** SciFive states the patient was "sent home from the hospital on 14/21," which is not a real date. The correct discharge date is 12/21, and the "stay" of 14 days has been erroneously merged into the date field. This reflects a failure to parse structured temporal information correctly.

- **Incorrect anatomical description — "widening of the chest":** The original text refers to "QRS widening," an electrocardiographic (ECG) finding indicating an abnormal electrical conduction pattern in the heart. SciFive misinterprets this as a physical finding of the chest widening, which is anatomically and clinically meaningless. This is a significant medical error.

- **“High potassium levels in the kidneys”:** Hyperkalemia refers to an elevated potassium level in the blood (serum), not specifically in the kidneys. Localizing it to the kidneys misrepresents the physiological condition and could mislead a patient’s family.

- **Contradictory and incoherent death narrative:** The model states “the person was released from the hospital without benefit, they went into cardiorespiratory arrest” and then “the body was resuscitated” followed by “the body was released.” These statements are internally contradictory — a resuscitated body that is subsequently released makes no clinical sense. This represents a failure to understand the sequential causality of the terminal clinical events.

- **Hallucinated cause of death — “burn in the ear, nose, and throat”:** The model states the death was “caused by a burn in the ear, nose, and throat.” There is no mention of any burn, ENT pathology, or ear/nose/throat condition anywhere in the original summary. This is a completely fabricated cause of death bearing no resemblance to the documented clinical events (Fournier’s gangrene and refractory septic shock).

- **Incorrect description of Fournier’s gangrene:** The model associates the gangrene with a “burn in the ear, nose, and throat,” which is entirely incorrect. Fournier’s gangrene is a rapidly spreading necrotizing infection of the genital and perineal region — not related to burns or ENT pathology in any way.

- **“Dobutamine” listed alongside diagnoses:** SciFive’s sentence structure merges the medication dobutamine with the list of diagnoses (“was given dobutamine [...] a swelling in the left thigh [...] and a condition called Fournier’s syndrome”), making it appear as though dobutamine is a diagnosis rather than a treatment. This structural confusion would mislead a lay reader.

- **Cardiac arrest and CPR details omitted:** The cardiac arrest with pulseless electrical activity (PEA), the 1-minute CPR episode, and the subsequent ventricular tachycardia (VT) with successful cardioversion are entirely absent from the output. These are major clinical events.

- **Norepinephrine dosing omitted:** The critical detail of norepinephrine being escalated to 1.5 mcg/kg/min as a vasopressor in refractory shock is not mentioned. This is a key indicator of the severity of the patient’s final deterioration.

- **Repetitive and redundant phrasing:** The output contains repeated, circular sentences such as “The family requested contact with the family” and “The body was released” appearing twice. These suggest decoding instability in the final generation stages.

Overall Assessment: SciFive produces the most fluent prose among the four models, but introduces critical factual errors including a hallucinated cause of death, misinterpretation of ECG findings, contradictory statements about patient outcome, and fabrication of non-existent conditions. Despite its relative readability, its factual unreliability makes it unsafe for clinical lay language conversion in its current fine-tuned state.

7.7 Comparative Summary

Table 2 provides a side-by-side comparison of model performance across key evaluation dimensions for this example.

Table 2: Qualitative comparison of model outputs on the representative discharge summary.

Criterion	BioBART	BioGPT	Flan-T5	SciFive
Factual accuracy	Poor	Poor	Poor	Partial
Death outcome correctly stated	No	Partial	No	No
Cause of death correct	No	No	No	No
No hallucinated content	No	No	No	No
Prose fluency	Moderate	Moderate	Poor	Good
Date accuracy	Poor	Partial	Partial	Poor
Medications preserved	Partial	Poor	Partial	Partial
No prompt leakage	Yes	Yes	No	Yes
Output completeness	Partial	Truncated	Degenerated	Partial
Appropriate tone	Partial	Poor	Poor	Partial

The results demonstrate that all four fine-tuned models fail to meet the requirements for safe and accurate lay language conversion on this complex case. The most critical failure mode shared across all models is **hallucination of clinical content** — particularly the cause of death. This highlights a fundamental limitation of current fine-tuning approaches: models optimized for fluency and surface-level simplification do not inherently acquire the factual grounding necessary for clinical document conversion. Future work should explore retrieval-augmented generation (RAG), constrained decoding strategies, or larger instruction-tuned foundation models as base architectures to address these shortcomings.

References

- Yuan, H., Yuan, Z., Gan, R., Zhang, J., Xie, Y., & Yu, S. (2022). BioBART: Pretraining and Evaluation of A Biomedical Generative Language Model. *arXiv preprint arXiv:2204.03905*. `https://arxiv.org/abs/2204.03905`

- Phan, L. N., Anibal, J. T., Tran, H., Chanana, S., Bahadıroğlu, E., Peltekian, A., & Altan-Bonnet, G. (2021). SciFive: a text-to-text transformer model for biomedical literature. *arXiv preprint arXiv:2106.03598*. `https://arxiv.org/abs/2106.03598`

- Luo, R., Sun, L., Xia, Y., Qin, T., Zhang, S., Poon, H., & Liu, T. Y. (2022). BioGPT: generative pre-trained transformer for biomedical text generation and mining. *Briefings in Bioinformatics*, 23(6). `https://arxiv.org/abs/2210.10341`

- Chung, H. W., Hou, L., Longpre, S., Zoph, B., Tay, Y., Fedus, W., ... & Wei, J. (2022). Scaling instruction-finetuned language models. *arXiv preprint arXiv:2210.11416*. `https://arxiv.org/abs/2210.11416`
- Zhang, J., Zhao, Y., Saleh, M., & Liu, P. J. (2020). PEGASUS: Pre-training with Extracted Gap-sentences for Abstractive Summarization. *International Conference on Machine Learning (ICML)*. `https://arxiv.org/abs/1912.08777`

A Indian Lay Language Dictionary (INDIAN_LAY_DICT)

The following table lists the medical terms and their corresponding lay language simplifications used in the preprocessing stage of the pipeline. These mappings were specifically curated to reflect common Indian English medical terminology.

Table 3: Medical Term to Lay Language Mapping

Medical Term	**Simplified Lay Language**
Conditions	
hypertension	high BP
diabetes mellitus	sugar disease (diabetes)
myocardial infarction	heart attack
acute renal failure	sudden kidney failure
chronic renal failure	long-term kidney problem
septicemia	blood infection (sepsis)
sepsis	serious blood infection
pneumonia	lung infection
tuberculosis	TB (tuberculosis)
pneumothorax	air trapped in chest
endocarditis	infection of heart valves
spondylodiscitis	spine bone infection
hypotension	low BP
pyrexia	fever
dyspnea	difficulty in breathing
edema	swelling
abscess	pus-filled swelling
crohn's disease	long-term bowel disease (Crohn's)
renal failure	kidney failure
cerebrovascular accident	brain stroke
anemia	low blood (anemia)
tachycardia	fast heartbeat
bradycardia	slow heartbeat
atrial fibrillation	irregular heartbeat
Procedures & Treatments	
hemodialysis	kidney dialysis
tracheostomy	breathing tube in neck
thoracotomy	chest surgery
pneumonectomy	removal of a lung

Medical Term	Simplified Lay Language
ileocolectomy	removal of part of bowel
anastomosis	surgical joining of bowel ends
debridement	surgical wound cleaning
arteriovenous fistula	dialysis access point on arm
intramuscular	given as a muscle injection
intravenous	given through a vein (drip)
percutaneous	through the skin
antibiotic therapy	antibiotic treatment
antibiotic	infection-fighting medicine
Lab & Medical Terms	
blood culture	blood test to find infection
procalcitonin	blood marker for infection
hemoglobin	blood count (Hb)
creatinine	kidney function marker
bilirubin	liver function marker
saturation	oxygen level in blood
afebrile	no fever
eupneic	breathing normally
acyanotic	no bluish discoloration
anicteric	no yellowing of skin/eyes
vesicular breath sounds	normal breathing sounds
Hospital / Admin	
discharge	sent home from hospital
outpatient clinic	OPD (Out Patient Department)
follow-up	return visit to doctor
ward	general hospital room
intensive care unit	ICU (serious care room)
sus	government health scheme
orally	by mouth
administer	give